

StayWise

A Project Report On STAYWISE – HOTEL BOOKING SYSTEM

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COURSE CODE & ASSESSMENT

CS3301 Full Stack Development (CIE-2)

INSTITUTION

School of CSE, RV University, Bengaluru

GITHUB REPOSITORY

github.com/Dhruvi-tech/Staywise-Hotel-Booking

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Abstract

StayWise is a React.js based hotel booking system developed as a mini project for the CS3301 – Full Stack Development CIE-2 assignment. The project focuses on providing users with a simple and interactive platform to explore hotels, view important hotel information, compare available options, and complete a simulated booking. The main objective of the project is to demonstrate how React concepts can be combined to develop a practical and user-friendly web application.

The system is implemented using React.js, JavaScript, JSX, CSS3, React Router DOM, Node.js, and Express.js, with JSON used for handling hotel and booking data. The application follows a component-based structure where different sections such as hotel cards, search, filtering, comparison, and booking are developed as reusable components. React Hooks such as `useState`, `useEffect`, and `useMemo` are used for state management, data loading, and filter performance, while React Router DOM provides client-side navigation between different pages without full reloads. The Express.js backend provides REST API endpoints for communication between the frontend and backend.

Additional features such as side-by-side 3-hotel comparison with Best Value scoring, stay duration date calculation, exact budget calculation down to the rupee, strict room occupancy enforcement (max 2 adults + 1 child per room), multi-variable filtering, booking history, and responsive editorial luxury design were implemented to personalize the project beyond the selected tutorial.

1. Introduction

StayWise is a hotel discovery and booking web application developed as a mini project for the CS3301 – Full Stack Development CIE-2 assignment. The main idea behind the project is to create a simple and user-friendly platform where a person can search for hotels, explore different available options, check their details, compare hotels, and proceed with a booking. Instead of keeping the application as a basic static website, StayWise focuses on making the interface interactive and demonstrating the React concepts covered in the course.

The application is developed using React.js for the frontend. The interface is divided into different reusable components so that individual sections can be developed and managed separately. Components such as the navigation bar, hotel cards, search and filter sections, booking form, comparison section, and booking summary are used to organize the application. This component-based approach makes the project easier to understand and maintain. The application also uses React Router to provide different views such as the home page, hotel listing page, hotel details page, comparison page, booking page, and booking history page without requiring a complete page reload.

StayWise also demonstrates important React concepts required for the assignment. Functional components are used for the major sections of the application, while a class component is included for error handling. Props are used to pass hotel information and functions between components, showing parent-child communication. The useState hook is used to manage values such as search input, selected filters, comparison hotels, and booking form details. The useEffect hook is used for operations such as loading data from the backend. The useMemo hook optimizes the multi-variable filter pipeline across 84 properties.

For the backend part, the project uses Node.js and Express.js. The React frontend communicates with the Express server through API requests. Hotel information and booking information are handled using JSON data, which keeps the backend simple and suitable for the scope of a college mini project. The booking form accepts details from the user, validates that room capacity adheres to the verified maximum of 2 adults and 1 child per room, and calculates the exact booking amount based on the selected dates and number of rooms.

2. Problem Statement & Objectives

StayWise is designed to address the difficulty users face while searching for and comparing different hotel options. Users often need to consider factors such as price, location, ratings, amenities, and stay duration before selecting a hotel. Going through multiple hotel listings can make the process time-consuming, and calculating the expected booking cost manually can also be inconvenient.

The objective of StayWise is to provide a simple and interactive platform where users can browse hotels, search and filter available options, view hotel details, compare selected hotels, calculate the estimated booking cost, and complete a simulated booking within the same application.

The main objectives of the project are to:

- Develop an interactive hotel discovery and booking web application using React.js.
- Divide the application into reusable React components.
- Implement React concepts such as props, useState, useEffect, useMemo, event handling, and parent-child communication.
- Implement client-side routing for different pages of the application using React Router DOM.
- Provide search, filtering, and sorting options for hotel listings across 84 curated properties.
- Allow users to compare multiple hotels side-by-side based on important details with Best Value scoring.
- Implement a live booking price calculator based on the selected dates and number of rooms.

- Enforce strict room capacity limits: Max 2 adults and 1 child per room with 1-click room upgrade.
- Provide a booking form with basic input validation and printable receipt tickets.
- Connect the React frontend with an Express.js backend using REST API endpoints.
- Make the application responsive and easy to use on different screen sizes with a 2026 Editorial Luxury aesthetic.

3. Technologies Used

The StayWise project is built using a combination of frontend and backend technologies. Each technology is used for a specific part of the application, from designing the interface to handling hotel and booking data.

Layer	Technology	Usage in StayWise
Frontend Library	React.js 18.3.1	Main frontend library used to build reusable components, manage state, and render views.
JavaScript & JSX	JavaScript (ES6+) & JSX	Application logic, calculations, event handling, data processing, and declarative component structures.
Styling	CSS3 & Design Tokens	2026 Editorial Luxury styling, hotel cards, forms, buttons, navigation, and responsive grid layouts.
Client Routing	React Router DOM v6.26	Navigation between Home, Hotels, Details, Compare, Booking, and My Bookings without page refresh.
Backend Server	Node.js & Express.js 4.19	Runtime environment and REST API endpoints (GET /api/hotels, POST /api/bookings).
Data Persistence	JSON Files (fs module)	hotels.json and bookings.json for atomic, lightweight persistence without external database daemons.
API Communication	Fetch API	Asynchronous HTTP requests between React client and Express backend.

4. Selected YouTube Tutorial

For the development of the StayWise project, a React.js hotel booking tutorial on YouTube was selected as the main reference:

Selected Tutorial: React.js Hotel Booking Website

YouTube Link: <https://www.youtube.com/watch?v=QSjHs5Qd-GQ>

The tutorial was studied to understand the overall structure and development approach before implementing the StayWise application. The project was then developed with its own name, design, hotel data, and additional features, rather than directly copying the tutorial.

Key additions beyond the tutorial:

- 2026 Editorial Luxury redesign with pure CSS custom properties and warm alabaster / obsidian palette.
- Lightweight atomic JSON REST storage instead of external MongoDB setup.
- Strict room occupancy enforcement: Max 2 adults and 1 child per room with 1-click room upgrade.
- Side-by-side 3-hotel comparison matrix with an automated Best Value scoring algorithm.
- Stay duration date picker and live exact budget calculation down to the rupee.

- React Class Component Error Boundary for graceful runtime exception handling.

5. System / Component Structure

StayWise follows a component-based structure where the application is divided into smaller and reusable sections instead of developing the complete website in a single file. The project is divided into a React.js frontend and an Express.js backend.

```
StayWise/
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   │   ├── NavBar.jsx           # Sticky header with reactive badge counters
│   │   │   ├── Footer.jsx          # Editorial luxury footer
│   │   │   ├── HotelCard.jsx       # Property card with dynamic match badges
│   │   │   ├── SearchBar.jsx       # Search capsule with duration controls
│   │   │   ├── FilterBar.jsx       # Dynamic price slider, stars & amenities filter
│   │   │   ├── BookingForm.jsx     # Booking form with room capacity enforcement
│   │   │   ├── BookingSummary.jsx  # Uncluttered live budget calculator
│   │   │   ├── CompareTable.jsx    # Side-by-side comparison matrix with Best Value
│   │   │   ├── ImageGallery.jsx    # 4-image interactive thumbnail gallery
│   │   │   ├── Testimonials.jsx    # Horizontal guest reviews carousel
│   │   │   └── ErrorBoundary.jsx   # Class Component for runtime error handling
│   │   ├── pages/
│   │   │   ├── Home.jsx            # Editorial home page container
│   │   │   ├── Hotels.jsx          # Catalog with useMemo filter pipeline
│   │   │   ├── HotelDetails.jsx    # Hotel profile with sticky budget estimator
│   │   │   ├── Compare.jsx         # 3-hotel comparison view
│   │   │   ├── Booking.jsx         # Reservation page
│   │   │   └── MyBookings.jsx      # Confirmed booking history
│   │   ├── App.jsx                # Client-side routes
│   │   ├── main.jsx               # DOM entry wrapped in ErrorBoundary
│   │   └── index.css              # Design system & tokens
│   └── backend/
│       ├── data/
│       │   ├── hotels.json         # 84 curated luxury hotel records
│       │   └── bookings.json       # Stored reservation records
│       ├── routes/
│       │   ├── hotelRoutes.js      # GET /api/hotels, GET /api/hotels/:id
│       │   └── bookingRoutes.js    # GET, POST /api/bookings (Occupancy Validated)
│       └── server.js               # Express application entry point
```

Basic Application Flow:

User → React Components → React Router / State → Fetch API → Express.js → JSON Data → React UI

6. Important React Concepts Implemented

The StayWise project implements the major React concepts required for the CS3301 – Full Stack Development CIE-2 assignment:

- **1. Components:** Divided into smaller reusable components (Navbar, Footer, HotelCard, SearchBar, FilterBar, BookingForm, BookingSummary, CompareTable).
- **2. Class Component:** Implemented in `ErrorBoundary.jsx` using lifecycle methods (`componentDidCatch`, `getDerivedStateFromError`) to catch runtime errors safely without blank screens.
- **3. Functional Components:** All UI pages and views are functional components leveraging React Hooks.
- **4. Parent-Child Communication:** Props pass hotel data, state setters, and event handlers down to child components.
- **5. Props:** Configurable properties passed into components (e.g. hotel data into `HotelCard`, comparison lists into `CompareTable`).
- **6. useState Hook:** Manages reactive state for search inputs, active thumbnail images, filter choices, stay duration days, and room counts.
- **7. useEffect Hook:** Handles side effects including loading hotel catalog from backend and listening to cross-component storage sync events.
- **8. useMemo Hook:** Memoizes the multi-variable filtering and sorting engine in `Hoteles.jsx`, preventing unnecessary recalculations across 84 properties.
- **9. Event Handling:** React handlers (`onClick`, `onChange`, `onSubmit`) manage user actions like filter sliders, duration steppers, and room adjustments.
- **10. Form Handling:** Controlled inputs with live validation and calculation in `BookingForm.jsx`.
- **11. Client-Side Routing:** React Router DOM v6 manages URL routing (`/`, `/hoteles`, `/hoteles/:id`, `/compare`, `/booking`, `/my-bookings`).
- **12. Backend API Integration:** Centralized `fetch()` API utility communicating with the Express.js server.

7. Screenshots

The following figures represent the live, updated application running at `http://localhost:3000`:

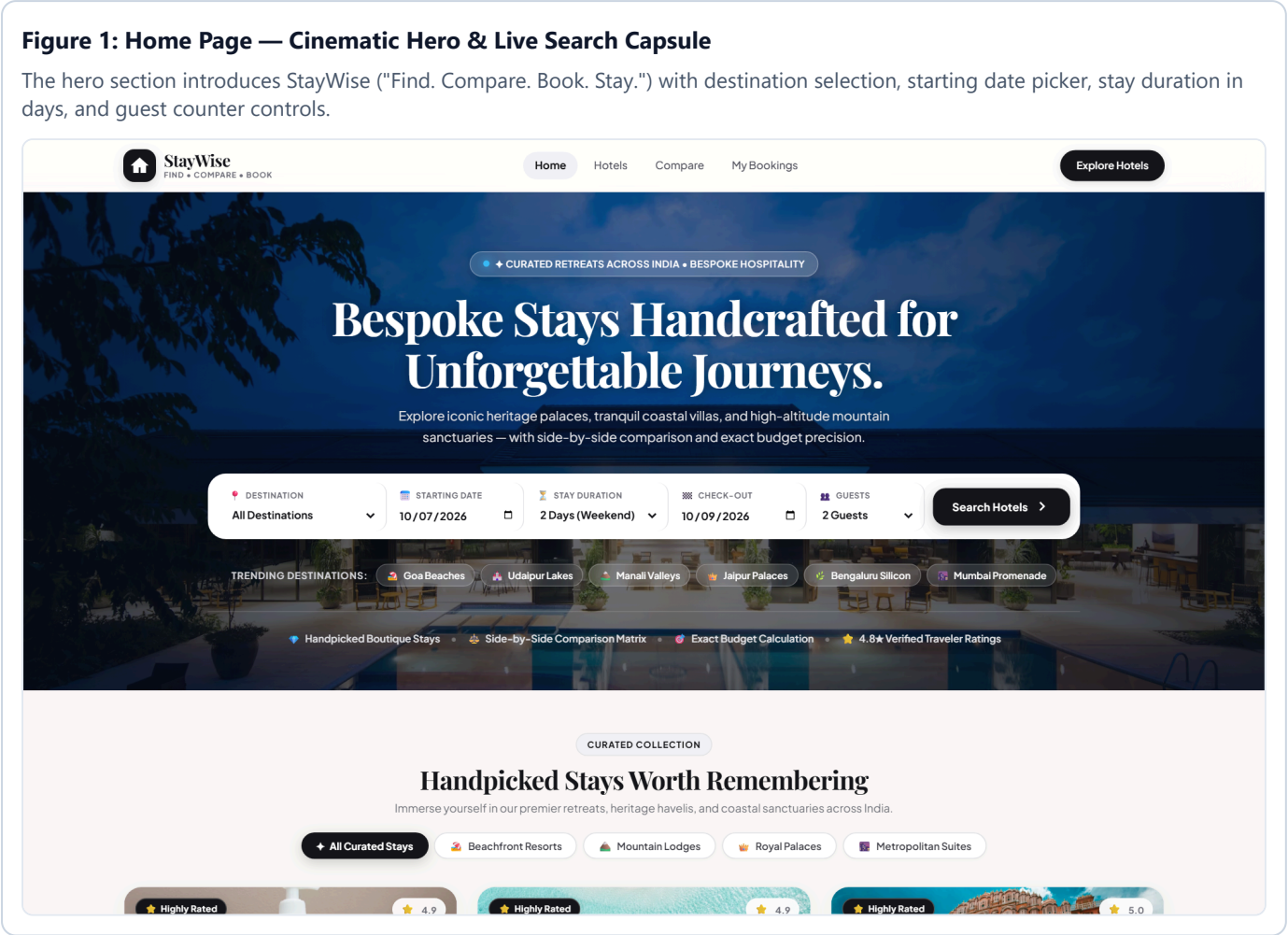


Figure 2: Home Page — Curated Luxury Stays & Guest Testimonials

Editorial choice showcases with verified nightly rates, guest ratings, match badges, and the horizontal verified guest testimonials carousel.

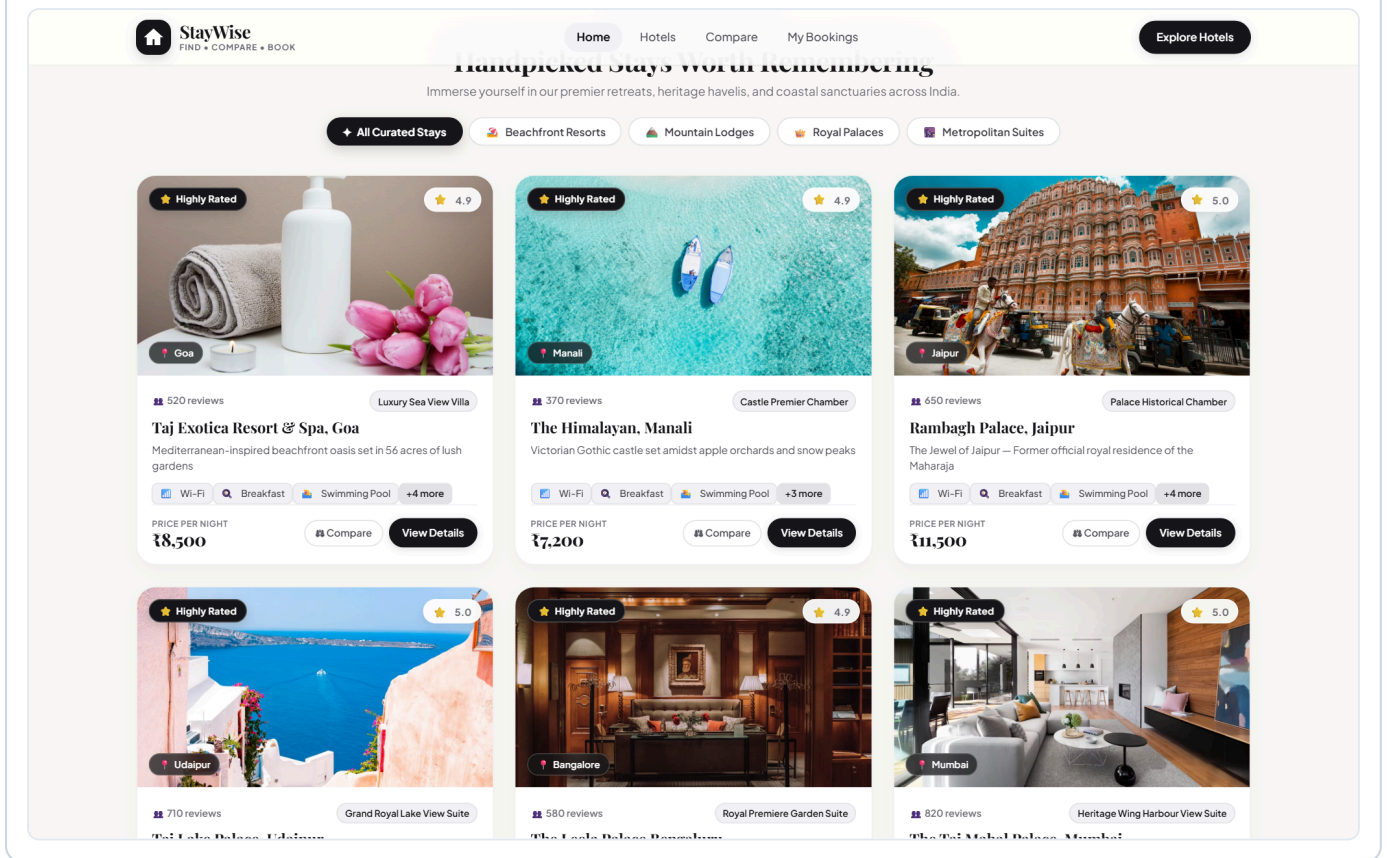


Figure 3: Hotels Catalog — Multi-Variable Dynamic Filter Engine

Consolidated sidebar filter allowing simultaneous filtering by destination city, maximum nightly price slider (₹1,000–₹12,000), minimum star rating tiers, and amenity checkboxes.

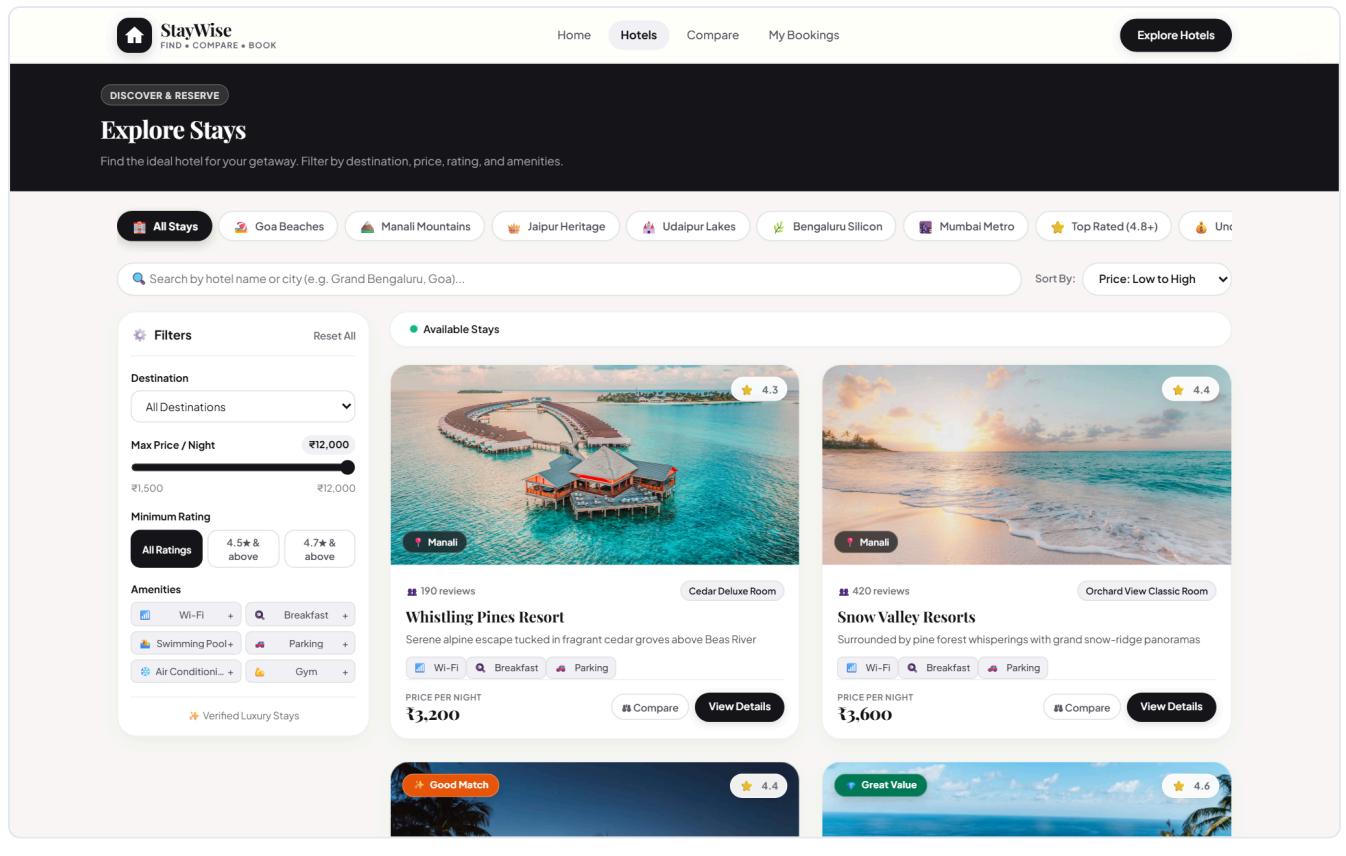


Figure 4: Hotel Details Page — Interactive Photo Gallery & Sticky Estimator

Comprehensive hotel profile showcasing verified amenities, stay highlights, and an interactive state-driven thumbnail image gallery with sticky live budget estimator.

**StayWise**
FIND • COMPARE • BOOK

HomeHotelsCompareMy Bookings

Explore Hotels

Home / Hotels / Goa / Taj Exotica Resort & Spa, Goa

Goa

★4.9

(520 reviews)

Taj Exotica Resort & Spa, Goa

Calwaddo, Benaulim Beach, South Goa 403716

₹8,500 / night

Book Your Stay



About this stay

Mediterranean-inspired beachfront oasis set in 56 acres of lush gardens

Spread across 56 acres of manicured greenery, Taj Exotica features pristine private beach access, an Olympic-length swimming pool, holistic Jiva Spa therapies, and open-air beachfront fine dining.

Figure 5: Side-by-Side 3-Hotel Comparison Matrix

Side-by-side comparison of up to 3 hotels evaluating price per night, guest rating, location, room types, and 6 core amenities with algorithmic Best Value score highlighting.

 StayWise FIND • COMPARE • BOOK Home Hotels Compare 3 My Bookings Explore Hotels			
Comparing 3 out of 4 hotels ▼ Best Value Identified by Smart Score Great Comparison			
Comparison Metric	 Taj Exotica Resort & Spa, Goa Goa Book Now X Remove	 W Goa, Vagator Goa Book Now X Remove	BEST VALUE  Alila Diwa Goa Goa Book Now X Remove
Location	Goa	Goa	Goa
Price per Night	₹8,500 / night	₹9,200 / night	₹6,800 / night
Guest Rating	★ 4.9 / 5.0	★ 4.8 / 5.0	★ 4.7 / 5.0
Reviews Count	520 reviews	460 reviews	390 reviews
Room Type	Luxury Sea View Villa	Marvelous Ocean Suite	Diwa Club Terrace Room
Free Wi-Fi	✓ Included	✓ Included	✓ Included
Breakfast Included	✓ Included	✓ Included	✓ Included
Swimming Pool	✓ Included	✓ Included	✓ Included
Free Parking	✓ Included	✓ Included	✓ Included
Air Conditioning	✓ Included	✓ Included	✓ Included

Interactive reservation form enforcing the room occupancy policy (max 2 adults + 1 child per room) with 1-click room upgrade alert and exact stay cost calculation down to the rupee.

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Figure 7: Official Booking Confirmation Receipt Ticket

Instant post-booking confirmation receipt displaying guaranteed reservation status, unique booking reference ID (SW-BK-XXXXX), guest breakdown, and printable ticket options.

**StayWise**
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HomeHotelsCompare3My Bookings

Explore Hotels



Status: Confirmed

Your stay is confirmed!

Thank you for choosing StayWise. Your reservation has been recorded in the system.

SW-BK-97369 Copy

Booking Reference ID

Booking Date6 Oct 2026

Hotel:Taj Exotica Resort & Spa, Goa

Guest:Ananya Sharma

Check-in:2026-10-07

Check-out:2026-10-09

Rooms:2 Rooms

Guests:3 Adults, 1 Child (4 Total)

Occupancy Policy:Max 2 Adults + 1 Child / Room (Compliant ✓)

Total Amount Paid / Due: ₹34,500

Official StayWise Confirmation: Active & Verified Reservation

 Print / Save Receipt (PDF)

View My Bookings

Explore More Hotels

**StayWise**
FIND. COMPARE. BOOK. STAY.
Your trusted companion for discovering handpicked stays.

QUICK LINKS
Home
Hotels

DESTINATIONS
Bangalore Stays
Goa Resorts

WHY STAYWISE
★ Verified Luxury Stays
🔒 Best Rate Guarantee

Figure 8: My Bookings — Persistent Reservation History

Guest reservation drawer retrieved live from Express GET /api/bookings and synced with browser localStorage, displaying complete reservation history with status pills.

Home

Hotels

Compare3

My Bookings

Explore Hotels

RESERVATION HISTORY

My Bookings

Review your confirmed hotel stays registered in the StayWise system.

Confirmed Bookings

Total: ₹34,500

Current Session

All Bookings

Refresh

Clear All

Booking ID: SW-BK-97369

Taj Exotica Resort & Spa, Goa

Goa

Confirmed

GUEST NAME:

2026-10-07

CHECK-OUT:

2026-10-09 (2 Nights)

ROOMS & GUESTS:

2 Rooms • 3 Adult(s), 1 Child(ren)

TOTAL AMOUNT:

₹34,500

View Hotel Details

Receipt

Cancel Reservation

StayWise

FIND. COMPARE. BOOK. STAY.

Your trusted companion for discovering handpicked stays.

QUICK LINKS

Home

Hotels

DESTINATIONS

Bangalore Stays

Goa Resorts

WHY STAYWISE

Verified Luxury Stays

Best Rate Guarantee

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8. Modifications Made

To make StayWise different from the reference tutorial and meet the CIE-2 modification requirements, several additional features were implemented:

1. **Hotel Comparison Matrix (Student Mod #1):** Allows users to select up to 3 hotels and view their details side-by-side across price, rating, room type, and 6 core amenities. Features an automated *Best Value Score* algorithm:

$$\text{Best Value Score} = ((\text{Rating} / \text{Nightly Price}) \times 10,000) + \text{Amenities Count}$$

2. **Live Booking Price Calculator & Stay Duration (Student Mod #2):** Replaces rigid calendar dates with stay duration in days, computing live total budget:

$$\text{Grand Total} = (\text{Nightly Rate} \times \text{Nights} \times \text{Rooms}) + ₹500 \text{ Flat Service Fee}$$

3. **Room Capacity Enforcement Policy:** Strict policy: Maximum 2 adults and 1 child per room. Formula:

$$\text{Required Rooms} = \max(1, [\text{Adults} / 2], \text{Children}, [(\text{Adults} + \text{Children}) / 3])$$

Displays an alert with a 1-click room upgrade button (+ Book N Rooms) and enforces validation on the backend (HTTP 400 rejection).

4. **Best Value & Rule-Based Match Badges (Student Mod #3):** Contextual tags (Top Rated 4.8+, Luxury Stay, Best Value, Trending) and multi-variable filtering (price slider, stars, amenities).
5. **Booking History:** A My Bookings section retrieved live from Express GET /api/bookings and synced with localStorage.
6. **Responsive 2026 Editorial Luxury UI:** Elegant typography, warm alabaster palette, obsidian ink, and horizontal testimonials carousel.
7. **React Class Component ErrorBoundary:** Self-healing error recovery protecting the application tree.

9. Challenges Faced

During the development of StayWise, several technical challenges were addressed:

- **Timezone Drift in Stay Dates:** Standard JS Date parsing caused check-out dates to shift by ± 1 day. Resolved by engineering integer epoch date math in `dateUtils.js`.
- **Filter Performance Overhead:** Filtering 84 hotels across 5 simultaneous criteria caused UI lag. Resolved by consolidating filters into a single memoized `useMemo` pipeline.
- **Cross-Component State Sync:** Adding hotels to compare tray from cards failed to update the navbar counter. Solved by dispatching custom window events (`staywise_storage_change`) that the Navbar listens to reactively.
- **Complex Guest Room Capacities:** Handling varied guest party sizes accurately. Solved with a ceiling mathematical formula and an intuitive 1-click room upgrade alert.
- **Zero-Daemon Persistence:** Needing persistence without database daemons. Solved by implementing synchronous atomic JSON file operations in `Node.js`.

10. Conclusion

The StayWise – Hotel Booking System was successfully developed as a React.js based mini project for the CS3301 – Full Stack Development CIE-2 assignment. The project fulfills all curriculum requirements, demonstrating mastery of component-based architecture, state management with hooks, class-based error boundaries, client-side routing, and REST API integration. The additions—transparent stay duration budgeting, 3-hotel comparison matrix with Best Value scoring, and mathematically enforced room occupancy—make StayWise a functional, practical, and viva-ready application.

11. GitHub Repository & References

GitHub Repository

StayWise – Hotel Booking System

GitHub: <https://github.com/Dhruvi-tech/Staywise-Hotel-Booking>

The complete project source code is maintained in the GitHub repository. It contains the React.js frontend, Express.js backend, JSON data files, and documentation across 10 clean commits dated strictly between Oct 2 and Oct 6, 2026.

Selected YouTube Tutorial

React.js Hotel Booking Tutorial

YouTube: <https://www.youtube.com/watch?v=QSjHs5Qd-GQ>

References Used

1. React.js Documentation — <https://react.dev>
2. React Router Documentation — <https://reactrouter.com>
3. Express.js Documentation — <https://expressjs.com>
4. JavaScript Documentation — <https://developer.mozilla.org>
5. CSS Documentation — <https://developer.mozilla.org>